Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 (Generated practice paper - Set 20)

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1 . 1** Describe **three** effects of noise on human health other than hearing damage.

**[3 marks]**

**1** .....

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**2** .....

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**3** .....

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**0 1 . 2** Explain how noise can damage non-living objects.

**[3 marks]**

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**0 1 . 3** Explain how noise can reduce the breeding success of birds.

**[2 marks]**

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**0 1 . 4** Explain why military sonar is a threat to cetaceans.

**[2 marks]**

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**0 2 . 1** Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

**[1 mark]**

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

**0 2 . 2** Explain how polymerising agents make an oil spill easier to clean up.

**[2 marks]**

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**0 2 . 3** Explain why deoxygenation is a more common problem in water than in air.

**[2 marks]**

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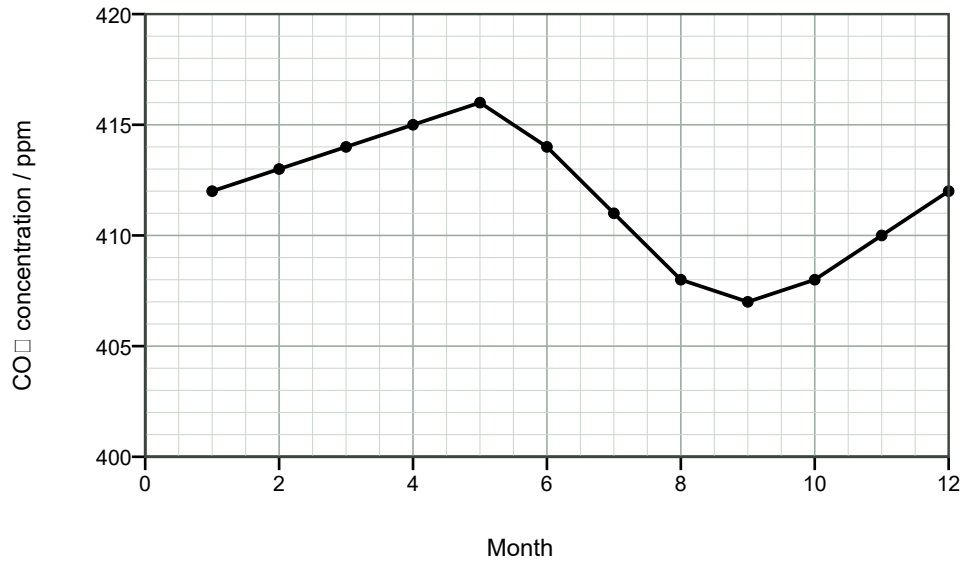
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**0 3** Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

**Table 1**

| Gas | Percentage by volume / % |
|---|---|
| Nitrogen | 78 |
| Oxygen | 21 |
| Argon and other rare gases | 0.9 |
| Carbon dioxide | 0.04 |
| Ozone | 0.000007 |

**Figure 1**




**0 3 . 1** Describe and explain the pattern shown in Figure 1.

**[4 marks]**

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**0 3 . 2** Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

**[2 marks]**

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**0 3 . 3** Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

**[2 marks]**

**1** .....

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**2** .....

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**0 3 . 4** Name the two layers of the atmosphere most affected by human activities.

**[2 marks]**

**1** .....

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**2** .....

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**0 4 . 1** Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

**[3 marks]**

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**0 4 . 2** Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

**[3 marks]**

**0 4 . 3** Describe the locational constraints of **two** named renewable energy resources.

**[4 marks]**

**10**

**0 5** Table 1 shows the efficiency and cost of four types of solar photovoltaic (PV) cell.

**Table 1**

| PV cell type | Efficiency / % | Relative cost per $m^2$ |
|---|---|---|
| Amorphous silicon (thin film) | 8 | 1 |
| Polycrystalline silicon | 16 | 2 |
| Monocrystalline silicon with anti-reflective surface | 22 | 3 |
| Multi-junction cell | 40 | 25 |

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**0 5 . 1** Describe how a photovoltaic cell generates electricity.

**[2 marks]**

**0 5 . 2** Explain how anti-reflective surfaces and multi-junction cells increase efficiency.

**[3 marks]**

**0 5 . 3** Use Table 1 to suggest which cell type would be chosen for a large solar farm on cheap desert land and which for a satellite. Justify your choices.

**[3 marks]**

**0 5 . 4** Explain why solar PV output is intermittent **and** unpredictable.

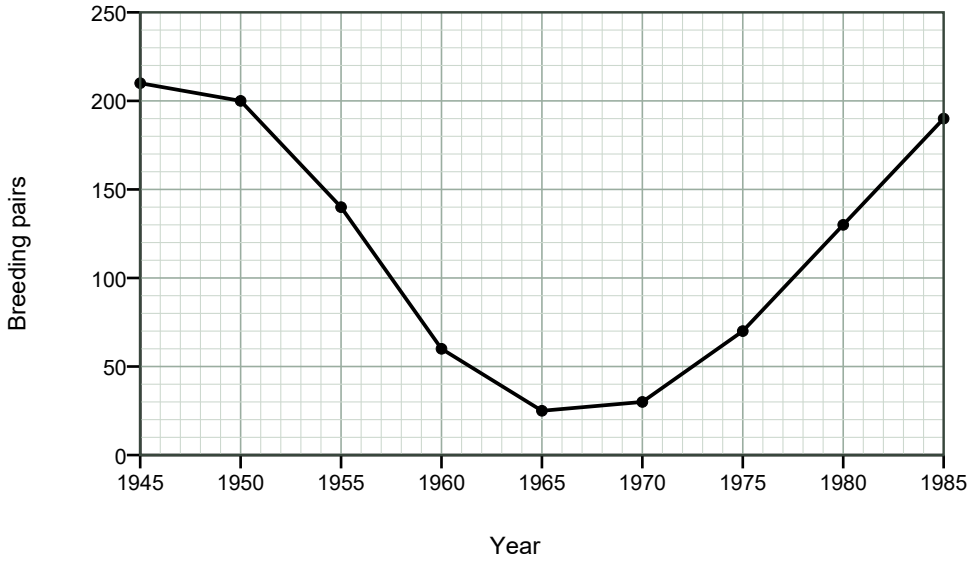
**[2 marks]**

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**0 6** Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

**Figure 1**




**0 6 . 1** Describe the trend shown in Figure 1.

**[2 marks]**

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**0 6 . 2** Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

**[2 marks]**

**Answer** ..... %

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**0 6 . 3** DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

**[4 marks]**

**0 6 . 4** Suggest why the recovery after 1970 was slow.

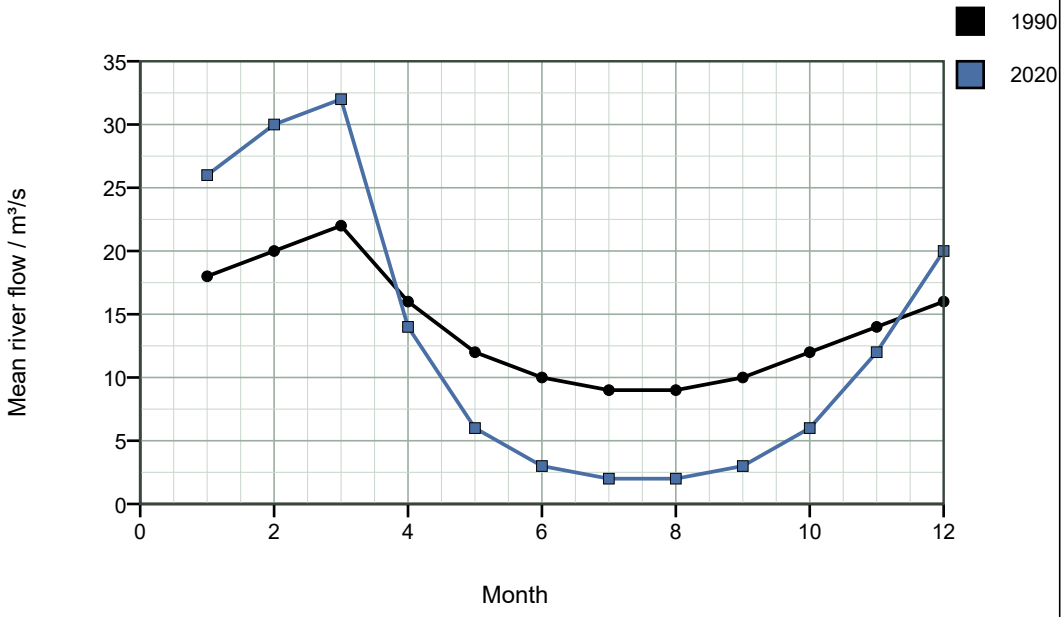
**[2 marks]**

**10**

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**0 7** Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

**Figure 1**




**0 7 . 1** Compare the river flow in 1990 and 2020.

**[3 marks]**

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**0 7 . 2** Explain how the shrinking of the glacier explains the differences.

**[3 marks]**

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**0 7 . 3** Suggest **two** problems the change in flow causes for people living downstream.

**[2 marks]**

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**2** .....

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**0 7 . 4** Explain why a glacier can be extending down its valley yet still be losing ice volume.

**[2 marks]**

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**0 8** A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

**Table 1**

| Location | Mean wind speed / m $s^{-1}$ | Standard deviation / m $s^{-1}$ |
|---|---|---|
| Woodland | 1.2 | 0.4 |
| Clearing | 3.5 | 1.1 |

**0 8 . 1** Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

**[3 marks]**

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**0 8 . 2** Explain what the student should do if it is not possible to show that the data are normally distributed.

**[1 mark]**

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**0 8 . 3** The test gave p = 0.02. Explain what this means.

**[2 marks]**

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**0 8 . 4** Explain how the standard deviations in Table 1 support the conclusion that the means are different.

**[2 marks]**

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**0 8 . 5** Explain why a statistically significant result does not prove that trees cause the lower wind speed.

**[2 marks]**

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**0 9 . 1** Give **three** advantages of incinerating domestic waste.

**[3 marks]**

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**2** .....

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**3** .....

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**0 9 . 2** Explain how each of the following features of a modern incinerator reduces pollution.

**[4 marks]**

**High combustion temperature with rapid cooling of gases** .....

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**Water spray scrubber** .....

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**Removal of wet / compostable waste** .....

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**Removal of recyclable waste** .....

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**0 9 . 3** Explain why the ash from waste incinerators needs careful disposal.

**[2 marks]**

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**0 9 . 4** State **one** reason why incineration is expensive.

**[1 mark]**

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**10**

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**1 0 . 1** Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

**[9 marks]**

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**1 0 . 2** State what is meant by **bioaccumulation**.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

**11 . 1** Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

**[25 marks]**

**OR**

**11 . 2** Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**